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Work Power & Energy

JEE Main Physics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	123 questions
Numerical	42 questions
Answer key	at the end

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MCQ (Single Correct Answer) 123 questions

Q1. JEE Main 2026 (Online) 6th April Evening Shift

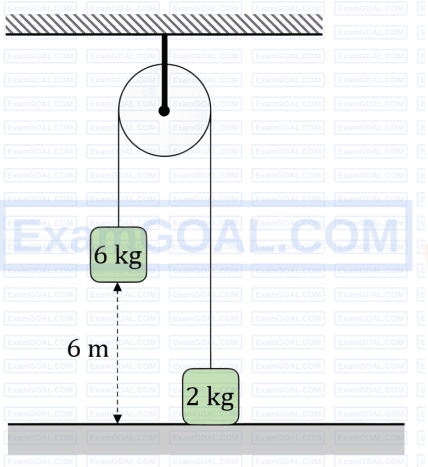
A body of mass 1 kg moves along a straight line with a velocity $v = 2x^2$. The work done by the body during displacement from $x = 0$ to 5 m is ____ J.

(A) 0 (B) 250 (C) 1250 (D) 1000

Open solution

Q2. JEE Main 2026 (Online) 6th April Evening Shift

The velocity at which 6 kg mass (shown in figure) strikes the ground when it is released from a height of 6 m above the ground is ____ $\mathrm{m/s}$. Assume pulley is massless and string is light and inextensible. (Take $\mathrm{g} = 10\ \mathrm{m/s^2}$)



(A) 7.74 (B) 7.20 (C) 6.55 (D) 4.50

Open solution

Q3. JEE Main 2026 (Online) 6th April Morning Shift

The rain drop of mass 1 g , starts with zero velocity from a height of 1 km . It hits the ground with a speed of $5\ \mathrm{m/s}$. The work done by the unknown resistive force is ____ J.

(take $\mathrm{g} = 10\ \mathrm{m/s^2}$)

(A) -8.75 (B) -8.35 (C) -9.55 (D) -9.98

Open solution

Q4. JEE Main 2026 (Online) 5th April Evening Shift

A mass of 1 kg is kept on an inclined plane with 30° inclination with respect to horizontal plane and it is at rest initially. Then the whole assembly is moved up with constant velocity of $4\ \mathrm{m/s}$. The work done by the frictional force in time 2 s is ____ J. (Take $g = 10\ \mathrm{m/s^2}$)

(A) 20 (B) 25 (C) 30 (D) 10

Open solution

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Q5. JEE Main 2026 (Online) 2nd April Evening Shift

Open solution

A spherical ball of mass 2 kg falls from a height of 10 m and is brought to rest after penetrating 10 cm into sand.

The average force exerted by sand on the ball is _____ N.

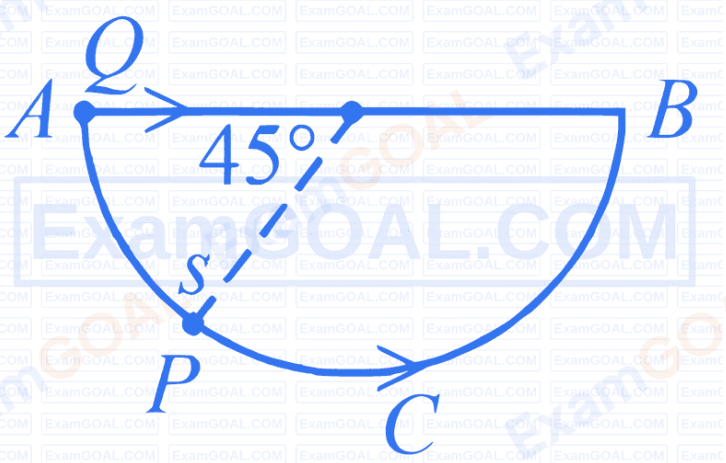
(Take $g = 10\ \mathrm{m/s^2}$)

(A) 1980 (B) 2020 (C) 2000 (D) 1000

Q6. JEE Main 2026 (Online) 23rd January Evening Shift

Open solution

A bead P sliding on a frictionless semi-circular string (ACB) and it is at point S at $t = 0$ and at this instant the horizontal component of its velocity is v. Another bead Q of the same mass as P is ejected from point A at $t = 0$ along the horizontal string AB, with the speed v, friction between the beads and the respective strings may be neglected in both cases. Let t_P and t_Q be the respective times taken by beads P and Q to reach the point B, then the relation between t_P and t_Q is



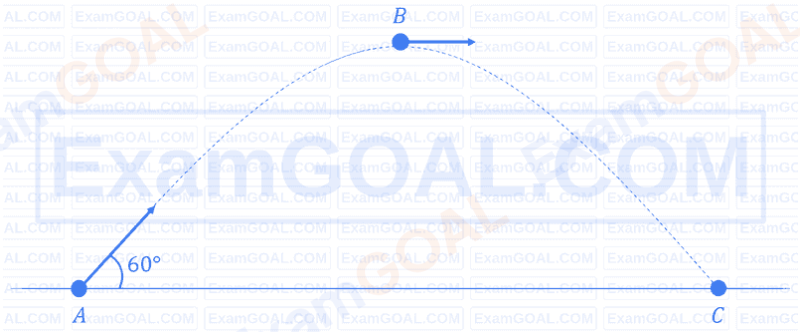
(A) $t_P < t_Q$ (B) $t_P = t_Q$ (C) $t_P > 1.25t_Q$ (D) $t_P > t_Q$

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Q7. JEE Main 2026 (Online) 23rd January Morning Shift

Open solution

An object is projected with kinetic energy K from a point A at an angle 60° with the horizontal. The ratio of the difference in kinetic energies at points B and C to that at point A (see figure), in the absence of air friction is :

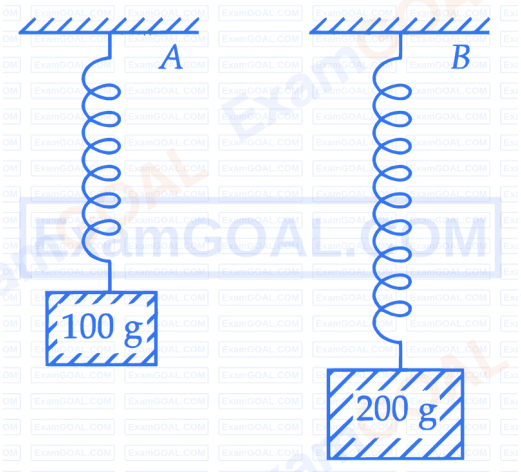


(A) $3:4$ (B) $1:4$ (C) $2:3$ (D) $1:2$

Q8. JEE Main 2026 (Online) 23rd January Morning Shift

Open solution

Two blocks with masses 100 g and 200 g are attached to the ends of springs A and B as shown in figure. The energy stored in A is E. The energy stored in B, when spring constants k_A, k_B of A and B, respectively satisfy the relation $4k_A = 3k_B$, is :



(A) $4E$ (B) $2E$ (C) $\frac{4}{3}E$ (D) $3E$

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Q9. JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

Given below are two statements :

Statement I : An object moves from position r_1 to position r_2 under a conservative force field $\vec{F}$. The work done by the force is $W = -\int_{r_1}^{r_2} \vec{F} \cdot \overrightarrow{dr}$.

Statement II : Any object moving from one location to another location can follow infinite number of paths. Therefore, the amount of work done by the object changes with the path it follows for a conservative force.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are false

(D) Both Statement I and Statement II are true

Q10. JEE Main 2026 (Online) 21st January Evening Shift

Open solution

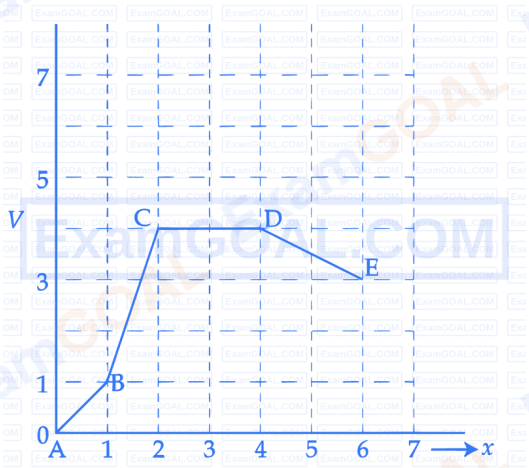
A body of mass 2 kg is moving along *x*-direction such that its displacement as function of time is given by $x(t) = \alpha t^2 + \beta t + \gamma$ m, where $\alpha = 1\ m/s^2$, $\beta = 1\ m/s$ and $\gamma = 1\ m$. The work done on the body during the time interval $t = 2\ s$ to $t = 3\ s$, is _____ J.

(A) 42 (B) 24 (C) 49 (D) 12

Q11. JEE Main 2026 (Online) 21st January Morning Shift

Open solution

Potential energy (V) versus distance (x) is given by the graph. Rank various regions as per the magnitudes of the force (F) acting on a particle from high to low.



(A) $F_{BC} > F_{AB} > F_{DE} > F_{CD}$

(B) $F_{BC} > F_{CD} > F_{DE} > F_{AB}$

(C) $F_{CD} > F_{AB} > F_{BC} > F_{DE}$

(D) $F_{CD} > F_{DE} > F_{AB} > F_{BC}$

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Q12. JEE Main 2025 (Online) 7th April Evening Shift

Which one of the following forces cannot be expressed in terms of potential energy?

(A) Frictional force
(B) Coulomb's force
(C) Restoring force
(D) Gravitational force

Open solution

Q13. JEE Main 2025 (Online) 7th April Morning Shift

An object of mass 1000 g experiences a time dependent force $\vec{F} = \left(2t\hat{i} + 3t^2\hat{j}\right)N$. The power generated by the force at time t is:

(A) $\left(2t^2 + 18t^3\right)W$ (B) $\left(3t^3 + 5t^5\right)W$ (C) $\left(2t^3 + 3t^5\right)W$ (D) $\left(2t^2 + 3t^3\right)W$

Open solution

Q14. JEE Main 2025 (Online) 4th April Evening Shift

A block of mass 25 kg is pulled along a horizontal surface by a force at an angle 45° with the horizontal. The friction coefficient between the block and the surface is 0.25 . The block travels at a uniform velocity. The work done by the applied force during a displacement of 5 m of the block is :

(A) 490 J (B) 970 J (C) 735 J (D) 245 J

Open solution

Q15. JEE Main 2025 (Online) 3rd April Morning Shift

A particle is released from height S above the surface of the earth. At certain height its kinetic energy is three times its potential energy. The height from the surface of the earth and the speed of the particle at that instant are respectively.

(A) $\frac{S}{4}, \frac{3gS}{2}$
(B) $\frac{S}{2}, \frac{3gS}{2}$
(C) $\frac{S}{4}, \sqrt{\frac{3gS}{2}}$
(D) $\frac{S}{2}, \sqrt{\frac{3gS}{2}}$

Open solution

Q16. JEE Main 2025 (Online) 29th January Evening Shift

A sand dropper drops sand of mass m(t) on a conveyer belt at a rate proportional to the square root of speed (v) of the belt, i.e., $\frac{dm}{dt} \propto \sqrt{v}$. If P is the power delivered to run the belt at constant speed then which of the following relationship is true?

(A) $P \propto \sqrt{v}$ (B) $P \propto v$ (C) $P^2 \propto v^3$ (D) $P^2 \propto v^5$

Open solution

Q17. JEE Main 2025 (Online) 28th January Evening Shift

A body of mass 4 kg is placed on a plane at a point P having coordinates $(3, 4)\mathrm{m}$. Under the action of force $\overrightarrow{F} = (2\hat{i} + 3\hat{j})\mathrm{N}$, it moves to a new point Q having coordinates $(6, 10)\mathrm{m}$ in 4 sec . The average power and instantaneous power at the end of 4 sec are in the ratio of :

(A) 4 : 3 (B) 13 : 6 (C) 1 : 2 (D) 6 : 13

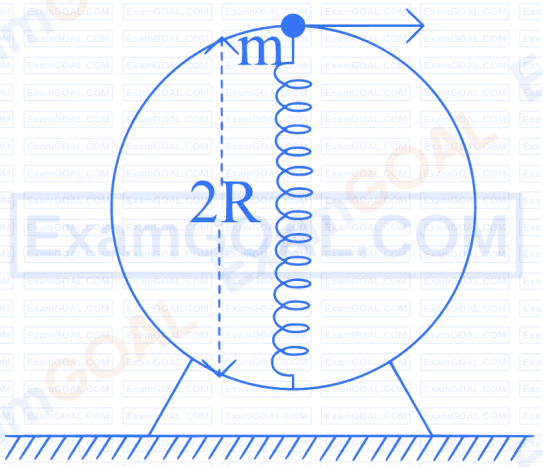
Open solution

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Q18. JEE Main 2025 (Online) 28th January Morning Shift

Open solution

A bead of mass ' m ' slides without friction on the wall of a vertical circular hoop of radius ' R ' as shown in figure. The bead moves under the combined action of gravity and a massless spring (k) attached to the bottom of the hoop. The equilibrium length of the spring is ' R '. If the bead is released from top of the hoop with (negligible) zero initial speed, velocity of bead, when the length of spring becomes ' R ', would be (spring constant is ' k ', g is acceleration due to gravity)



(A) $\sqrt{2Rg+\frac{kR^2}{m}}$

(B) $\sqrt{3Rg+\frac{kR^2}{m}}$

(C) $\sqrt{2Rg+\frac{4kR^2}{m}}$

(D) $2\sqrt{gR+\frac{kR^2}{m}}$

Q19. JEE Main 2025 (Online) 28th January Morning Shift

Open solution

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason $\mathbf{R}$

Assertion A: In a central force field, the work done is independent of the path chosen.

Reason R: Every force encountered in mechanics does not have an associated potential energy.

In the light of the above statements, choose the most appropriate answer from the options given below

(A) $\mathbf{A}$ is false but $\mathbf{R}$ is true

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

(C) $\mathbf{A}$ is true but $\mathbf{R}$ is false

(D) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

Q20. JEE Main 2025 (Online) 24th January Morning Shift

Open solution

A force $F=\alpha+\beta x^2$ acts on an object in the x -direction. The work done by the force is 5 J when the object is displaced by 1 m . If the constant $\alpha=1\ N$ then β will be

(A) $15\ N/m^2$

(B) $10\ N/m^2$

(C) $12\ N/m^2$

(D) $8\ N/m^2$

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Q21. JEE Main 2025 (Online) 23rd January Evening Shift

A ball having kinetic energy KE, is projected at an angle of 60° from the horizontal. What will be the kinetic energy of ball at the highest point of its flight?

(A) $\frac{(\mathrm{KE})}{2}$ (B) $\frac{(\mathrm{KE})}{8}$ (C) $\frac{(\mathrm{KE})}{4}$ (D) $\frac{(\mathrm{KE})}{16}$

Open solution

Q22. JEE Main 2025 (Online) 22nd January Evening Shift

A force $\vec{\mathrm{F}} = 2\hat{i} + \mathrm{b}\hat{j} + \hat{k}$ is applied on a particle and it undergoes a displacement $\hat{i} - 2\hat{j} - \hat{k}$. What will be the value of b, if work done on the particle is zero.

(A) $\frac{1}{3}$ (B) $\frac{1}{2}$ (C) 0 (D) 2

Open solution

Q23. JEE Main 2024 (Online) 9th April Morning Shift

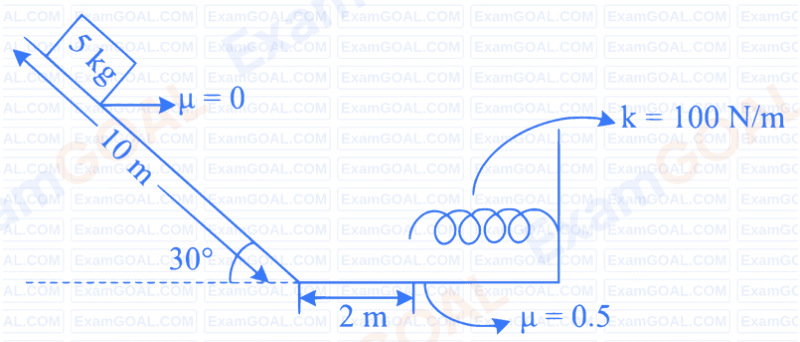
A particle of mass m moves on a straight line with its velocity increasing with distance according to the equation $v = \alpha\sqrt{x}$, where α is a constant. The total work done by all the forces applied on the particle during its displacement from $x = 0$ to $x = \mathrm{d}$, will be :

(A) $\frac{\mathrm{m}}{2\alpha^2\,\mathrm{d}}$ (B) $\frac{\mathrm{md}}{2\alpha^2}$

(C) $\frac{\mathrm{m}\alpha^2\,\mathrm{d}}{2}$ (D) $2\,\mathrm{m}\alpha^2\,\mathrm{d}$

Open solution

Q24. JEE Main 2024 (Online) 8th April Evening Shift



A block is simply released from the top of an inclined plane as shown in the figure above. The maximum compression in the spring when the block hits the spring is :

(A) $\sqrt{6}\,\mathrm{m}$ (B) $\sqrt{5}\,\mathrm{m}$ (C) $1\,\mathrm{m}$ (D) $2\,\mathrm{m}$

Open solution

Q25. JEE Main 2024 (Online) 8th April Morning Shift

Three bodies A, B and C have equal kinetic energies and their masses are $400\,\mathrm{g}, 1.2\,\mathrm{kg}$ and $1.6\,\mathrm{kg}$ respectively. The ratio of their linear momenta is :

(A) $1 : \sqrt{3} : 2$ (B) $\sqrt{3} : \sqrt{2} : 1$ (C) $1 : \sqrt{3} : \sqrt{2}$ (D) $\sqrt{2} : \sqrt{3} : 1$

Open solution

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Q26. JEE Main 2024 (Online) 6th April Evening Shift

When kinetic energy of a body becomes 36 times of its original value, the percentage increase in the momentum of the body will be :

(A) 60% (B) 500% (C) 6% (D) 600%

Open solution

Q27. JEE Main 2024 (Online) 6th April Morning Shift

A bullet of mass $50\ \mathrm{g}$ is fired with a speed $100\ \mathrm{m/s}$ on a plywood and emerges with $40\ \mathrm{m/s}$. The percentage loss of kinetic energy is :

(A) 44% (B) 16% (C) 84% (D) 32%

Open solution

Q28. JEE Main 2024 (Online) 6th April Morning Shift

Four particles A, B, C, D of mass $\frac{m}{2}, m, 2m, 4m$, have same momentum, respectively. The particle with maximum kinetic energy is :

(A) B (B) C (C) D (D) A

Open solution

Q29. JEE Main 2024 (Online) 5th April Evening Shift

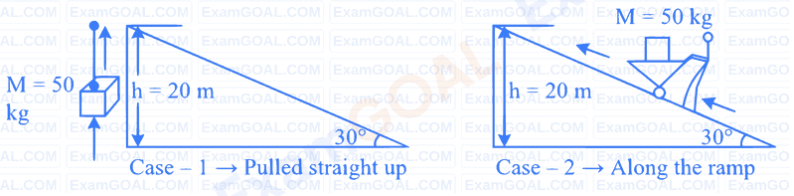
A body is moving unidirectionally under the influence of a constant power source. Its displacement in time t is proportional to :

(A) $t^{2/3}$ (B) $t^{3/2}$ (C) t (D) t^2

Open solution

Q30. JEE Main 2024 (Online) 5th April Morning Shift

A body of mass $50\ \mathrm{kg}$ is lifted to a height of $20\ \mathrm{m}$ from the ground in the two different ways as shown in the figures. The ratio of work done against the gravity in both the respective cases, will be :



(A) $2:1$ (B) $\sqrt{3}:2$ (C) $1:1$ (D) $1:2$

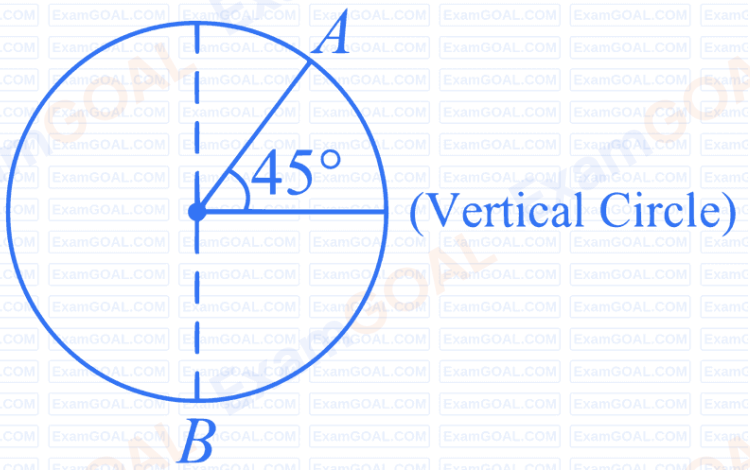
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Q31. JEE Main 2024 (Online) 4th April Evening Shift

Open solution

A body of m kg slides from rest along the curve of vertical circle from point A to B in frictionless path. The velocity of the body at B is:



(given, $R = 14\,\text{m}, g = 10\,\text{m/s}^2$ and $\sqrt{2} = 1.4$)

(A) 10.6 m/s (B) 19.8 m/s (C) 16.7 m/s (D) 21.9 m/s

Q32. JEE Main 2024 (Online) 4th April Morning Shift

Open solution

If a rubber ball falls from a height h and rebounds upto the height of $h/2$. The percentage loss of total energy of the initial system as well as velocity of ball before it strikes the ground, respectively, are :

(A) $50\%, \sqrt{2gh}$ (B) $50\%, \sqrt{gh}$

(C) $50\%, \sqrt{\frac{gh}{2}}$ (D) $40\%, \sqrt{2gh}$

Q33. JEE Main 2024 (Online) 31st January Evening Shift

Open solution

A body of mass 2 kg begins to move under the action of a time dependent force given by $\vec{F} = \left(6t\hat{i} + 6t^2\hat{j}\right)N$. The power developed by the force at the time t is given by:

(A) $\left(3t^3 + 6t^5\right)W$ (B) $\left(9t^5 + 6t^3\right)W$ (C) $\left(6t^4 + 9t^5\right)W$ (D) $\left(9t^3 + 6t^5\right)W$

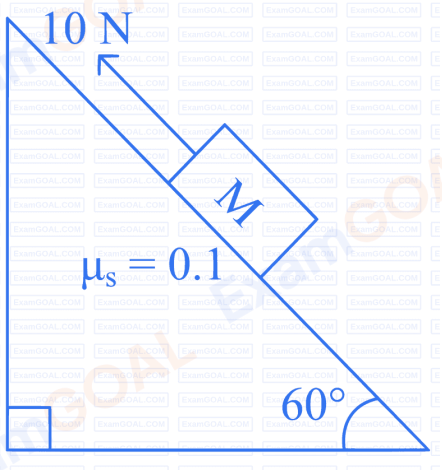
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Q34. JEE Main 2024 (Online) 30th January Evening Shift

Open solution

A block of mass $1\ \text{kg}$ is pushed up a surface inclined to horizontal at an angle of 60° by a force of $10\ \text{N}$ parallel to the inclined surface as shown in figure. When the block is pushed up by $10\ \text{m}$ along inclined surface, the work done against frictional force is :

$[g = 10\ \text{m/s}^2]$



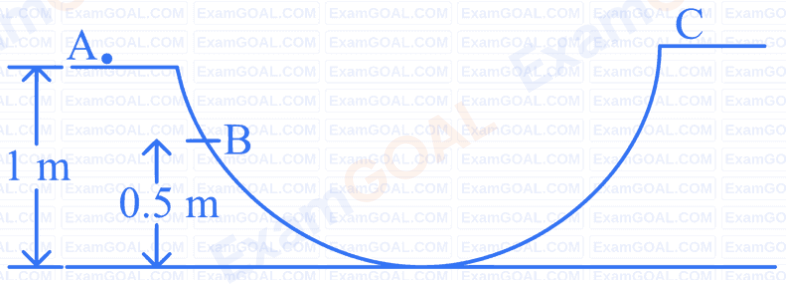
(A) $5\sqrt{3}$ J (B) 5 J (C) 5×10^3 J (D) 10 J

Q35. JEE Main 2024 (Online) 30th January Morning Shift

Open solution

A particle is placed at the point A of a frictionless track ABC as shown in figure. It is gently pushed towards right. The speed of the particle when it reaches the point B is :

(Take $g = 10\ \text{m/s}^2$).

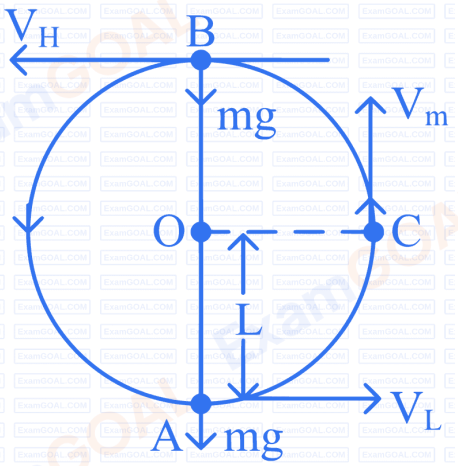


(A) $2\sqrt{10}\ \text{m/s}$ (B) $10\ \text{m/s}$
(C) $\sqrt{10}\ \text{m/s}$ (D) $20\ \text{m/s}$

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Q36. JEE Main 2024 (Online) 29th January Evening Shift

A bob of mass 'm' is suspended by a light string of length 'L'. It is imparted a minimum horizontal velocity at the lowest point A such that it just completes half circle reaching the top most position B. The ratio of kinetic energies $\frac{(K.E)_A}{(K.E)_B}$ is :



Open solution

(A) 5 : 1 (B) 3 : 2 (C) 1 : 5 (D) 2 : 5

Q37. JEE Main 2024 (Online) 29th January Morning Shift

A block of mass $100\ \mathrm{kg}$ slides over a distance of $10\ \mathrm{m}$ on a horizontal surface. If the co-efficient of friction between the surfaces is 0.4, then the work done against friction $(\mathrm{in}\ J)$ is :

(A) 3900 (B) 4500 (C) 4200 (D) 4000

Open solution

Q38. JEE Main 2024 (Online) 29th January Morning Shift

The potential energy function (in J) of a particle in a region of space is given as $U = (2x^2 + 3y^3 + 2z)$. Here x, y and z are in meter. The magnitude of x-component of force (in N) acting on the particle at point $P(1, 2, 3)\mathrm{m}$ is :

(A) 4 (B) 2 (C) 8 (D) 6

Open solution

Q39. JEE Main 2024 (Online) 27th January Evening Shift

A ball suspended by a thread swings in a vertical plane so that its magnitude of acceleration in the extreme position and lowest position are equal. The angle (θ) of thread deflection in the extreme position will be :

(A) $\tan^{-1}\left(\frac{1}{2}\right)$ (B) $2\tan^{-1}\left(\frac{1}{2}\right)$

(C) $2\tan^{-1}\left(\frac{1}{\sqrt{5}}\right)$ (D) $\tan^{-1}(\sqrt{2})$

Open solution

Q40. JEE Main 2024 (Online) 27th January Evening Shift

A bullet is fired into a fixed target loses one third of its velocity after travelling $4\ \mathrm{cm}$. It penetrates further $\mathrm{D} \times 10^{-3}\ \mathrm{m}$ before coming to rest. The value of D is :

(A) 23 (B) 32 (C) 42 (D) 52

Open solution

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Q41. JEE Main 2023 (Online) 13th April Morning Shift

The ratio of powers of two motors is $\frac{3\sqrt{x}}{\sqrt{x}+1}$, that are capable of raising $300\ \mathrm{kg}$ water in 5 minutes and $50\ \mathrm{kg}$ water in 2 minutes respectively from a well of $100\ \mathrm{m}$ deep. The value of x will be

(A) 16 (B) 4 (C) 2 (D) 2.4

Open solution

Q42. JEE Main 2023 (Online) 13th April Morning Shift

Two bodies are having kinetic energies in the ratio 16 : 9. If they have same linear momentum, the ratio of their masses respectively is :

(A) $3:4$ (B) $4:3$ (C) $9:16$ (D) $16:9$

Open solution

Q43. JEE Main 2023 (Online) 12th April Morning Shift

Given below are two statements:

Statement I : A truck and a car moving with same kinetic energy are brought to rest by applying brakes which provide equal retarding forces. Both come to rest in equal distance.

Statement II : A car moving towards east takes a turn and moves towards north, the speed remains unchanged. The acceleration of the car is zero.

In the light of given statements, choose the most appropriate answer from the options given below

(A) Statement I is incorrect but Statement II is correct.

(B) Statement I is correct but Statement II is incorrect.

(C) Both Statement I and Statement II are correct.

(D) Both Statement I and Statement II are incorrect.

Open solution

Q44. JEE Main 2023 (Online) 8th April Evening Shift

A bullet of mass $0.1\ \mathrm{kg}$ moving horizontally with speed $400\ \mathrm{ms}^{-1}$ hits a wooden block of mass $3.9\ \mathrm{kg}$ kept on a horizontal rough surface. The bullet gets embedded into the block and moves $20\ \mathrm{m}$ before coming to rest. The coefficient of friction between the block and the surface is _____.

(Given $g = 10\ \mathrm{m/s^2}$)

(A) 0.65 (B) 0.25 (C) 0.50 (D) 0.90

Open solution

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Q45. JEE Main 2023 (Online) 29th January Evening Shift

Open solution

Identify the correct statements from the following :

A. Work done by a man in lifting a bucket out of a well by means of a rope tied to the bucket is negative.

B. Work done by gravitational force in lifting a bucket out of a well by a rope tied to the bucket is negative.

C. Work done by friction on a body sliding down an inclined plane is positive.

D. Work done by an applied force on a body moving on a rough horizontal plane with uniform velocity is zero.

E. Work done by the air resistance on an oscillating pendulum is negative.

Choose the correct answer from the options given below :

(A) A and C only (B) B and D only (C) B, D and E only (D) B and E only

Q46. JEE Main 2023 (Online) 29th January Morning Shift

Open solution

A stone is projected at angle 30° to the horizontal. The ratio of kinetic energy of the stone at point of projection to its kinetic energy at the highest point of flight will be -

(A) 1 : 4 (B) 1 : 2 (C) 4 : 3 (D) 4 : 1

Q47. JEE Main 2022 (Online) 29th July Morning Shift

Open solution

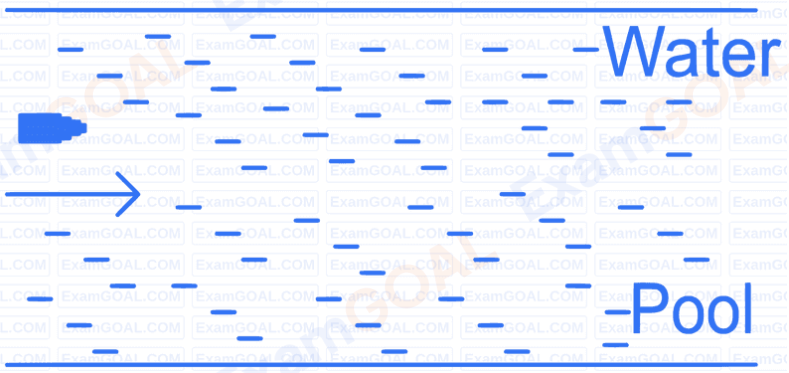
A ball is projected with kinetic energy E, at an angle of 60° to the horizontal. The kinetic energy of this ball at the highest point of its flight will become :

(A) Zero (B) $\frac{E}{2}$ (C) $\frac{E}{4}$ (D) E

Q48. JEE Main 2022 (Online) 28th July Evening Shift

Open solution

A bullet of mass $200\ \text{g}$ having initial kinetic energy $90\ \text{J}$ is shot inside a long swimming pool as shown in the figure. If its kinetic energy reduces to $40\ \text{J}$ within $1\ \text{s}$, the minimum length of the pool, the bullet has to travel so that it completely comes to rest is



(A) 45 m (B) 90 m (C) 125 m (D) 25 m

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Q49. JEE Main 2022 (Online) 27th July Morning Shift

Sand is being dropped from a stationary dropper at a rate of $0.5\,\mathrm{kgs}^{-1}$ on a conveyor belt moving with a velocity of $5\,\mathrm{ms}^{-1}$. The power needed to keep the belt moving with the same velocity will be :

(A) 1.25 W (B) 2.5 W (C) 6.25 W (D) 12.5 W

Open solution

Q50. JEE Main 2022 (Online) 26th July Morning Shift

As per the given figure, two blocks each of mass $250\,\mathrm{g}$ are connected to a spring of spring constant $2\,\mathrm{Nm}^{-1}$. If both are given velocity v in opposite directions, then maximum elongation of the spring is :



(A) $\frac{v}{2\sqrt{2}}$ (B) $\frac{v}{2}$ (C) $\frac{v}{4}$ (D) $\frac{v}{\sqrt{2}}$

Open solution

Q51. JEE Main 2022 (Online) 25th July Evening Shift

A bag of sand of mass 9.8 kg is suspended by a rope. A bullet of 200 g travelling with speed 10 ms $^{-1}$ gets embedded in it, then loss of kinetic energy will be :

(A) 4.9 J (B) 9.8 J (C) 14.7 J (D) 19.6 J

Open solution

Q52. JEE Main 2022 (Online) 25th July Morning Shift

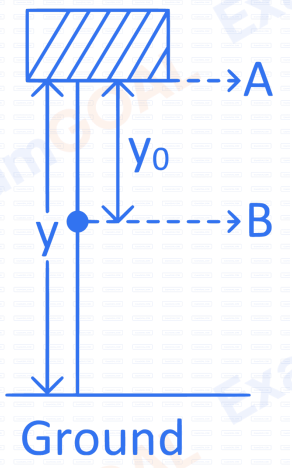
A body of mass $0.5\,\mathrm{kg}$ travels on straight line path with velocity $v = (3x^2 + 4)\mathrm{m/s}$. The net workdone by the force during its displacement from $x = 0$ to $x = 2\,\mathrm{m}$ is :

(A) 64 J (B) 60 J (C) 120 J (D) 128 J

Open solution

Q53. JEE Main 2022 (Online) 29th June Evening Shift

In the given figure, the block of mass m is dropped from the point 'A'. The expression for kinetic energy of block when it reaches point 'B' is



(A) $\frac{1}{2}mg{y_0}^2$ (B) $\frac{1}{2}mg\,y^2$ (C) $mg(y - y_0)$ (D) mgy_0

Open solution

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Q54. JEE Main 2022 (Online) 29th June Morning Shift

A particle of mass 500 gm is moving in a straight line with velocity $v = b x^{5/2}$. The work done by the net force during its displacement from x = 0 to x = 4 m is : (Take $b = 0.25\ m^{-3/2}\ s^{-1}$).

(A) 2 J (B) 4 J (C) 8 J (D) 16 J

Open solution

Q55. JEE Main 2022 (Online) 28th June Evening Shift

A block of mass 2 kg moving on a horizontal surface with speed of 4 ms^{-1} enters a rough surface ranging from x = 0.5 m to x = 1.5 m. The retarding force in this range of rough surface is related to distance by $F = -kx$ where $k = 12\ Nm^{-1}$. The speed of the block as it just crosses the rough surface will be :

(A) zero (B) 1.5 ms^{-1} (C) 2.0 ms^{-1} (D) 2.5 ms^{-1}

Open solution

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Q56. JEE Main 2022 (Online) 26th June Evening Shift

Open solution

Arrange the four graphs in descending order of total work done; where W_1, W_2, W_3 and W_4 are the work done corresponding to figure a, b, c and d respectively.

Figure - a

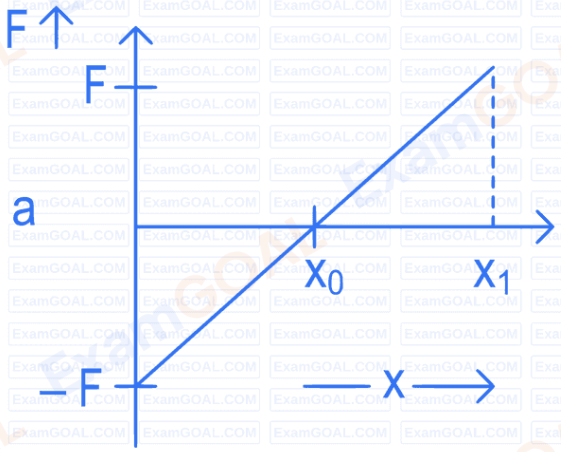


Figure - b

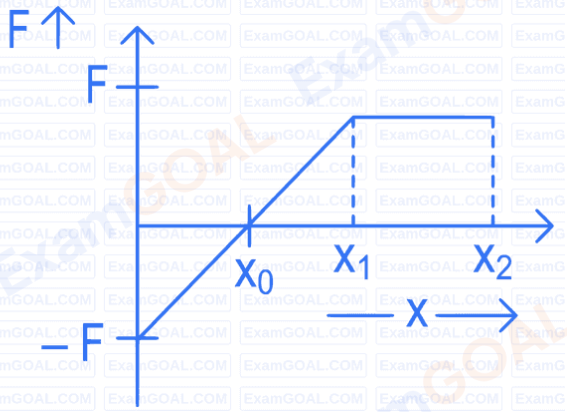
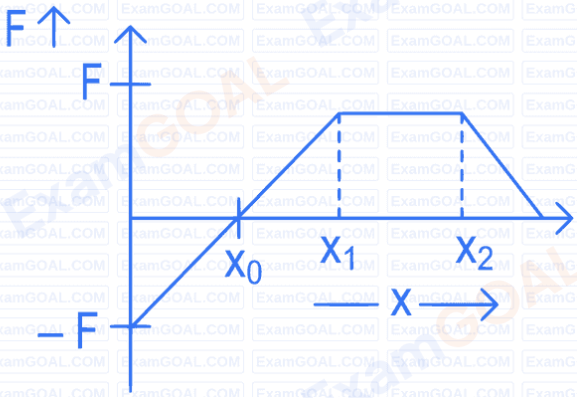
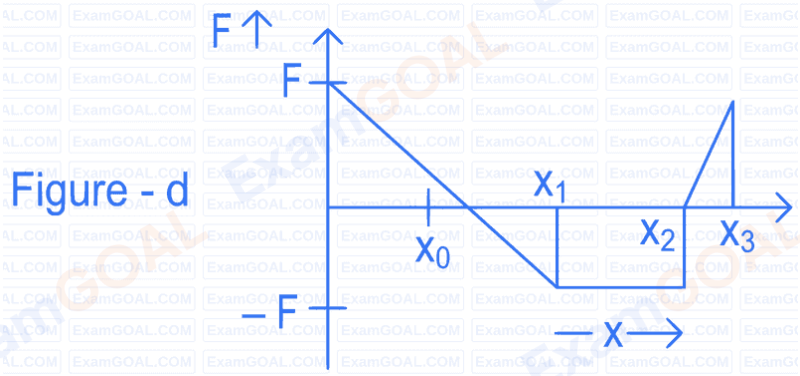


Figure - c



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(A) $W_3 > W_2 > W_1 > W_4$ (B) $W_3 > W_2 > W_4 > W_1$

(C) $W_2 > W_3 > W_4 > W_1$ (D) $W_2 > W_3 > W_1 > W_4$

Q57. JEE Main 2022 (Online) 24th June Evening Shift

Potential energy as a function of r is given by $U = \frac{A}{r^{10}} - \frac{B}{r^5}$, where r is the interatomic distance, A and B are positive constants. The equilibrium distance between the two atoms will be :

Open solution

(A) $\left(\frac{A}{B}\right)^{\frac{1}{5}}$ (B) $\left(\frac{B}{A}\right)^{\frac{1}{5}}$

(C) $\left(\frac{2A}{B}\right)^{\frac{1}{5}}$ (D) $\left(\frac{B}{2A}\right)^{\frac{1}{5}}$

Q58. JEE Main 2022 (Online) 24th June Morning Shift

A particle experiences a variable force $\overrightarrow{F} = \left(4x\hat{i} + 3y^2\hat{j}\right)$ in a horizontal x-y plane. Assume distance in meters and force is newton. If the particle moves from point (1, 2) to point (2, 3) in the x-y plane, then Kinetic Energy changes by :

Open solution

(A) 50.0 J (B) 12.5 J (C) 25.0 J (D) 0 J

Q59. JEE Main 2021 (Online) 1st September Evening Shift

A body of mass 'm' dropped from a height 'h' reaches the ground with a speed of $0.8\sqrt{gh}$. The value of work done by the air-friction is :

Open solution

(A) -0.68 mgh (B) mgh (C) 1.64 mgh (D) 0.64 mgh

Q60. JEE Main 2021 (Online) 31st August Evening Shift

A block moving horizontally on a smooth surface with a speed of 40 m/s splits into two parts with masses in the ratio of 1 : 2. If the smaller part moves at 60 m/s in the same direction, then the fractional change in kinetic energy is :-

Open solution

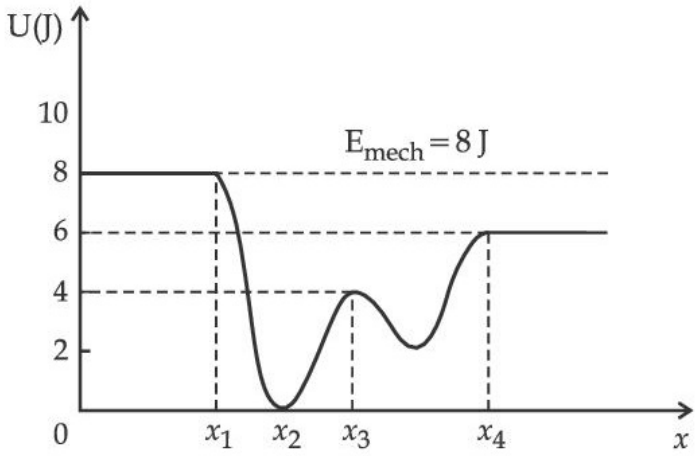
(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{1}{8}$ (D) $\frac{1}{4}$

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Q61. JEE Main 2021 (Online) 27th July Evening Shift

Open solution

Given below is the plot of a potential energy function U(x) for a system, in which a particle is in one dimensional motion, while a conservative force F(x) acts on it. Suppose that E_{mech} = 8 J, the incorrect statement for this system is :



[where K.E. = kinetic energy]

(A) at $x > x_4$ K.E. is constant throughout the region.

(B) at $x < x_1$, K.E. is smallest and the particle is moving at the slowest speed.

(C) at $x = x_2$, K.E. is greatest and the particle is moving at the fastest speed.

(D) at $x = x_3$, K.E. = 4 J.

Q62. JEE Main 2021 (Online) 27th July Evening Shift

Open solution

An automobile of mass 'm' accelerates starting from origin and initially at rest, while the engine supplies constant power P. The position is given as a function of time by :

(A) $\left(\frac{9P}{8m}\right)^{\frac{1}{2}} t^{\frac{3}{2}}$

(B) $\left(\frac{8P}{9m}\right)^{\frac{1}{2}} t^{\frac{2}{3}}$

(C) $\left(\frac{9m}{8P}\right)^{\frac{1}{2}} t^{\frac{3}{2}}$

(D) $\left(\frac{8P}{9m}\right)^{\frac{1}{2}} t^{\frac{3}{2}}$

Q63. JEE Main 2021 (Online) 22th July Evening Shift

Open solution

A porter lifts a heavy suitcase of mass 80 kg and at the destination lowers it down by a distance of 80 cm with a constant velocity. Calculate the work done by the porter in lowering the suitcase.

(take g = 9.8 ms^{-2})

(A) +627.2 J

(B) −62720.0 J

(C) −627.2 J

(D) 784.0 J

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Q64. JEE Main 2021 (Online) 20th July Evening Shift

A body at rest is moved along a horizontal straight line by a machine delivering a constant power. The distance moved by the body in time 't' is proportional to :

(A) $t^{\frac{3}{2}}$ (B) $t^{\frac{1}{2}}$ (C) $t^{\frac{1}{4}}$ (D) $t^{\frac{3}{4}}$

Open solution

Q65. JEE Main 2021 (Online) 18th March Morning Shift

A constant power delivering machine has towed a box, which was initially at rest, along a horizontal straight line. The distance moved by the box in time 't' is proportional to :-

(A) $t^{2/3}$ (B) $t^{3/2}$ (C) t (D) $t^{1/2}$

Open solution

Q66. JEE Main 2021 (Online) 17th March Morning Shift

A boy is rolling a 0.5 kg ball on the frictionless floor with the speed of 20 ms^{-1}. The ball gets deflected by an obstacle on the way. After deflection it moves with 5% of its initial kinetic energy. What is the speed of the ball now?

(A) 14.41 ms^{-1} (B) 19.0 ms^{-1} (C) 4.47 ms^{-1} (D) 1.00 ms^{-1}

Open solution

Q67. JEE Main 2020 (Online) 6th September Morning Slot

If the potential energy between two molecules is given by
U = $-\frac{A}{r^6}+\frac{B}{r^{12}}$,
then at equilibrium, separation between molecules, and the potential energy are :

(A) $\left(\frac{2B}{A}\right)^{1/6}, -\frac{A^2}{4B}$ (B) $\left(\frac{2B}{A}\right)^{1/6}, -\frac{A^2}{2B}$
(C) $\left(\frac{B}{A}\right)^{1/6}, 0$ (D) $\left(\frac{B}{2A}\right)^{1/6}, -\frac{A^2}{2B}$

Open solution

Q68. JEE Main 2020 (Online) 4th September Evening Slot

A person pushes a box on a rough horizontal platform surface. He applies a force of 200 N over a distance of 15 m. Thereafter, he gets progressively tired and his applied force reduces linearly with distance to 100 N. The total distance through which the box has been moved is 30 m. What is the work done by the person during the total movement of the box?

(A) 5690 J (B) 5250 J (C) 2780 J (D) 3280 J

Open solution

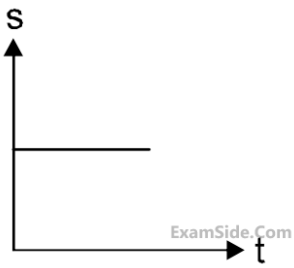
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Q69. JEE Main 2020 (Online) 3rd September Evening Slot

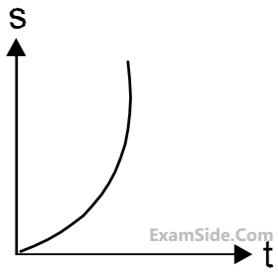
Open solution

A particle is moving unidirectionally on a horizontal plane under the action of a constant power supplying energy source. The displacement (s) - time (t) graph that describes the motion of the particle is (graphs are drawn schematically and are not to scale) :

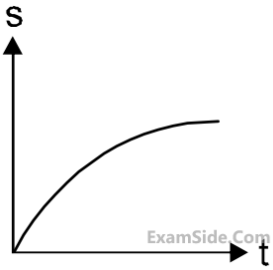
(A)



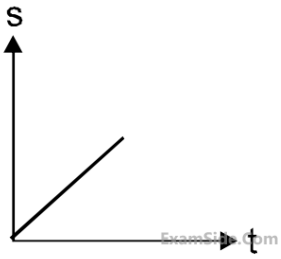
(B)



(C)



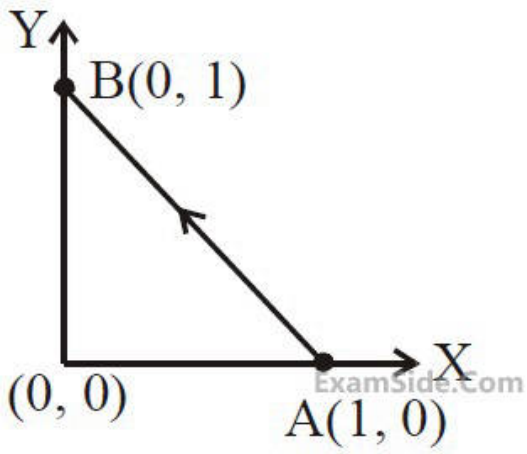
(D)



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Q70. JEE Main 2020 (Online) 9th January Morning Slot

Consider a force $\vec{F} = -x\hat{i} + y\hat{j}$. The work done by this force in moving a particle from point A(1, 0) to B(0, 1) along the line segment is : (all quantities are in SI units)



Open solution

(A) 2 (B) $\frac{1}{2}$ (C) 1 (D) $\frac{3}{2}$

Q71. JEE Main 2020 (Online) 7th January Evening Slot

An elevator in a building can carry a maximum of 10 persons, with the average mass of each person being 68 kg. The mass of the elevator itself is 920 kg and it moves with a constant speed of 3 m/s. The frictional force opposing the motion is 6000 N. If the elevator is moving up with its full capacity, the power delivered by the motor to the elevator (g = 10 m/s^2) must be at least :

Open solution

(A) 48000 W (B) 62360 W (C) 56300 W (D) 66000 W

Q72. JEE Main 2020 (Online) 7th January Morning Slot

A 60 HP electric motor lifts an elevator having a maximum total load capacity of 2000 kg. If the frictional force on the elevator is 4000 N, the speed of the elevator at full load is close to :
(1 HP = 746 W, g = 10 ms^{-2})

Open solution

(A) 1.5 ms^{-1} (B) 1.7 ms^{-1} (C) 2.0 ms^{-1} (D) 1.9 ms^{-1}

Q73. JEE Main 2019 (Online) 9th April Morning Slot

A uniform cable of mass 'M' and length 'L' is placed on a horizontal surface such that its (1/n)th part is hanging below the edge of the surface. To lift the hanging part of the cable upto the surface, the work done should be :

Open solution

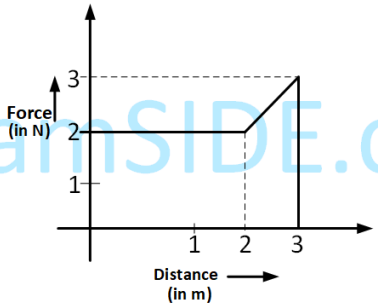
(A) $\frac{2MgL}{n^2}$ (B) nMgL (C) $\frac{MgL}{2n^2}$ (D) $\frac{MgL}{n^2}$

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Q74. JEE Main 2019 (Online) 8th April Morning Slot

A particle moves in one dimension from rest under the influence of a force that varies with the distance travelled by the particle as shown in the figure. The kinetic energy of the particle after it has travelled 3m is :

Open solution



ExamSIDE.com

(A) 6.5 J (B) 2.5 J (C) 5 J (D) 4 J

Q75. JEE Main 2019 (Online) 10th January Evening Slot

A particle which is experiencing a force, given by $\vec{F} = 3\hat{i} - 12\hat{j}$, undergoes a displacement of $\vec{d} = 4\vec{i}$ particle had a kinetic energy of 3 J at the beginning of the displacement, what is its kinetic energy at the end of the displacement ?

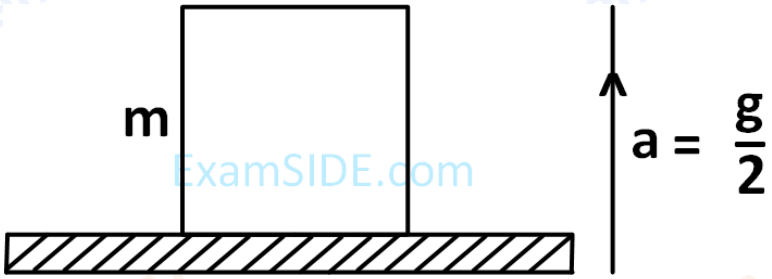
Open solution

(A) 9 J (B) 10 J (C) 12 J (D) 15 J

Q76. JEE Main 2019 (Online) 10th January Morning Slot

A block of mass m is kept on a platform which starts from rest with constant acceleration g/2 upward, as shown in figure. Work done by normal reaction on block in time t is -

Open solution



ExamSIDE.com

(A) $\frac{mg^2t^2}{8}$ (B) $\frac{3mg^2t^2}{8}$ (C) $-\frac{mg^2t^2}{8}$ (D) 0

Q77. JEE Main 2019 (Online) 9th January Evening Slot

A force acts on a 2 kg object so that its position is given as a function of time as $x = 3t^2 + 5$. What is the work done by this force in first 5 seconds ?

Open solution

(A) 850 J (B) 950 J (C) 875 J (D) 900 J

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Q78. JEE Main 2019 (Online) 9th January Morning Slot

A block of mass m, lying on a smooth horizontal surface, is attached to a spring (of negligible mass) of spring constant k. The other end of the spring is fixed, as shown in the figure. The block is initially at rest in its equilibrium position. If now the block is pulled with a constant force F, the maximum speed of the block is :



Open solution

(A) $\frac{2F}{\sqrt{mk}}$ (B) $\frac{F}{\pi\sqrt{mk}}$ (C) $\frac{\pi F}{\sqrt{mk}}$ (D) $\frac{F}{\sqrt{mk}}$

Q79. JEE Main 2018 (Online) 16th April Morning Slot

A body of mass m starts moving from rest along x-axis so that its velocity varies as $v = a\sqrt{s}$ where a is a constant and s is the distance covered by the body. The total work done by all the forces acting on the body in the first t seconds after the start of the motion is :

Open solution

(A) $\frac{1}{8}$ m a^4 t^2 (B) 8 m a^4 t^2 (C) 4 m a^4 t^2 (D) $\frac{1}{4}$ m a^4 t^2

Q80. JEE Main 2018 (Online) 16th April Morning Slot

Two particles of the same mass m are moving in circular orbits because of force, given by $F(r) = \frac{-16}{r} - r^3$

Open solution

The first particle is at a distance r = 1, and the second, at r = 4. The best estimate for the ratio of kinetic energies of the first and the second particle is closest to :

(A) 6×10^{-2} (B) 3×10^{-3} (C) 10^{-1} (D) 6×10^{2}

Q81. JEE Main 2018 (Offline)

A particle is moving in a circular path of radius a under the action of an attractive potential $U = -\frac{k}{2r^2}$. Its total energy is:

Open solution

(A) $-\frac{3}{2}\frac{k}{a^2}$ (B) Zero (C) $-\frac{k}{4a^2}$ (D) $\frac{k}{2a^2}$

Q82. JEE Main 2017 (Online) 8th April Morning Slot

An object is dropped from a height h from the ground. Every time it hits the ground it loses 50% of its kinetic energy. The total distance covered as t $\rightarrow \infty$ is :

Open solution

(A) 3h (B) ∞ (C) $\frac{5}{3}$h (D) $\frac{8}{3}$h

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Q83. JEE Main 2017 (Offline)

A time dependent force F = 6t acts on a particle of mass 1 kg. If the particle starts from rest, the work done by the force during the first 1 sec. will be:

Open solution

(A) 18 J (B) 4.5 J (C) 22 J (D) 9 J

Q84. JEE Main 2016 (Online) 10th April Morning Slot

A particle of mass M is moving in a circle of fixed radius R in such a way that its centripetal acceleration at time t is given by $n^2 R t^2$ where n is a constant. The power delivered to the particle by the force acting on it, is :

Open solution

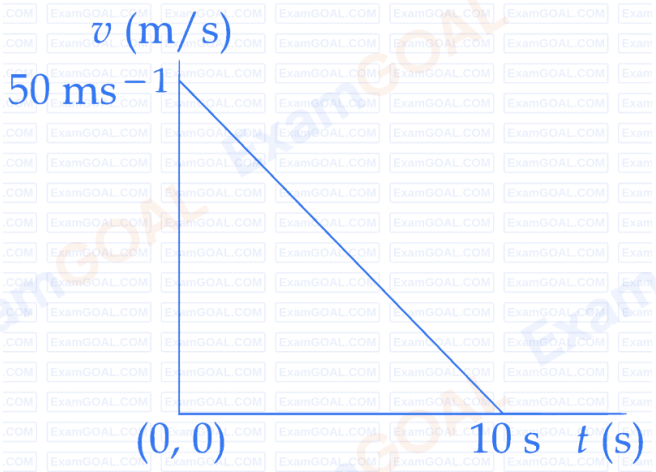
(A) $M n^2 R^2 t$ (B) $M n R^2 t$

(C) $M n R^2 t^2$ (D) $\frac{1}{2} M n^2 R^2 t^2$

Q85. JEE Main 2016 (Online) 10th April Morning Slot

Velocity-time graph for a body of mass 10 kg is shown in figure. Work-done on the body in first two seconds of the motion is :

Open solution



(A) 12000 J (B) − 12000 J (C) − 4500 J (D) − 9300 J

Q86. JEE Main 2016 (Online) 9th April Morning Slot

A car of weight W is on an inclined road that rises by 100 m over a distance of 1 km and applies a constant frictional force $\frac{W}{20}$ on the car. While moving uphill on the road at a speed of 10 ms^{-1}, the car needs power P. If it needs power $\frac{P}{2}$ while moving downhill at speed v then value of v is :

Open solution

(A) 20 ms^{-1} (B) 15 ms^{-1} (C) 10 ms^{-1} (D) 5 ms^{-1}

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Q87. JEE Main 2016 (Offline)

Open solution

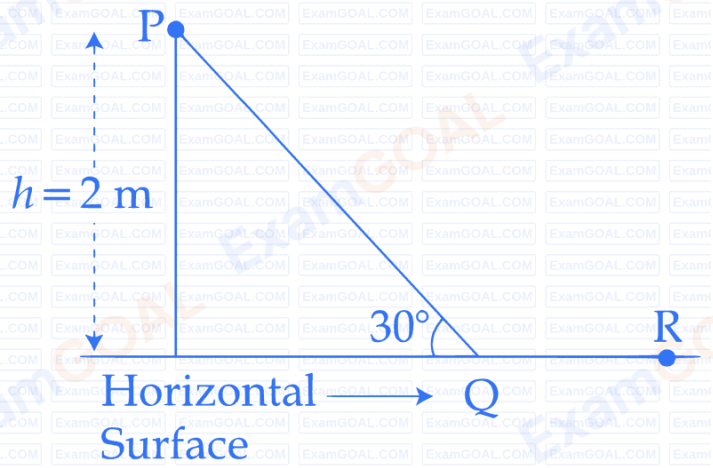
A person trying to lose weight by burning fat lifts a mass of $10\ kg$ upto a height of $1\ m$ 1000 times. Assume that the potential energy lost each time he lowers the mass is dissipated. How much fat will he use up considering the work done only when the weight is lifted up? Fat supplies $3.8 \times 10^7 J$ of energy per kg which is converted to mechanical energy with a 20% efficiency rate. Take $g = 9.8\, ms^{-2}$:

(A) $9.89 \times 10^{-3}\ kg$ (B) $12.89 \times 10^{-3}\, kg$ (C) $2.45 \times 10^{-3}\ kg$ (D) $6.45 \times 10^{-3}\ kg$

Q88. JEE Main 2016 (Offline)

Open solution

A point particle of mass m, moves along the uniformly rough track PQR as shown in the figure. The coefficient of friction, between the particle and the rough track equals μ. The particle is released, from rest from the point P and it comes to rest at point R. The energies, lost by the ball, over the parts, PQ and QR, of the track, are equal to each other , and no energy is lost when particle changes direction from PQ to QR.
The value of the coefficient of friction μ and the distance $x\ (= QR)$, are, respectively close to:



(A) 0.29 and $3.5\, m$ (B) 0.29 and $6.5\, m$ (C) 0.2 and $6.5\, m$ (D) 0.2 and $3.5\, m$

Q89. JEE Main 2015 (Online) 11th April Morning Slot

Open solution

A particle is moving in a circle of radius r under the action of a force F = αr^2 which is directed towards center of the circle. Total mechanical energy (kinetic energy + potential energy) of the particle is (take potential energy = 0 for r = 0) :

(A) $\alpha\, r^3$ (B) $\frac{1}{2}\alpha\, r^3$

(C) $\frac{4}{3}\alpha\, r^3$ (D) $\frac{5}{6}\alpha\, r^3$

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Q90. JEE Main 2015 (Online) 10th April Morning Slot

A block of mass m = 0.1 kg is connected to a spring of unknown spring constant k. It is compressed to a distance x from its equilibrium position and released from rest.

After approaching half the distance $\left(\frac{x}{2}\right)$ from equilibrium position, it hits another block and comes to rest momentarily, while the other block moves with a velocity 3 ms^{-1} . The total initial energy of the spring is :

(A) 1.5 J (B) 0.6 J (C) 0.3 J (D) 0.8 J

Open solution

Q91. JEE Main 2014 (Online) 19th April Morning Slot

A bullet loses $\left(\frac{1}{n}\right)^{th}$ of its velocity passing through one plank. The number of such planks that are required to stop the bullet can be :

(A) $\frac{n^2}{2n-1}$ (B) $\frac{2n^2}{n-1}$ (C) Infinite (D) n

Open solution

Q92. JEE Main 2014 (Online) 12th April Morning Slot

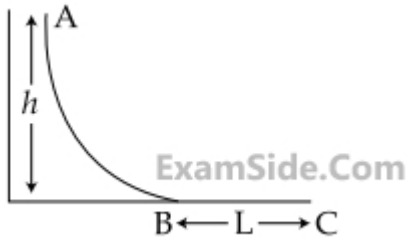
A spring of unstretched length l has a mass m with one end fixed to a rigid support. Assuming spring to be made of a uniform wire, the kinetic energy possessed by it if its free end is pulled with uniform velocity v is :

(A) $\frac{1}{2}$ m v^2 (B) m v^2 (C) $\frac{1}{3}$ m v^2 (D) $\frac{1}{6}$ m v^2

Open solution

Q93. JEE Main 2014 (Online) 11th April Morning Slot

A small ball of mass m starts at a point A with speed v_0 and moves along a frictionless track AB as shown. The track BC has coefficient of friction μ. The ball comes to stop at C after travelling a distance L which is :



(A) $\frac{2h}{\mu}+\frac{v_0^2}{2\mu g}$ (B) $\frac{h}{\mu}+\frac{v_0^2}{2\mu g}$

(C) $\frac{h}{2\mu}+\frac{v_0^2}{\mu g}$ (D) $\frac{h}{2\mu}+\frac{v_0^2}{2\mu g}$

Open solution

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Q94. JEE Main 2014 (Online) 11th April Morning Slot

The average mass of rain drops is 3.0×10^{-5} kg and their average terminal velocity is 9 m/s. Calculate the energy transferred by rain to each square meter of the surface at a place which receives 100 cm of rain in a year.

Open solution

(A) 3.5×10^5J (B) 4.05×10^4J (C) 3.0×10^5 J (D) 9.0×10^4 J

Q95. JEE Main 2014 (Offline)

When a rubber-band is stretched by a distance x, it exerts restoring force of magnitude $F = ax + bx^2$ where a and b are constants. The work done in stretching the unstretched rubber-band by L is :

Open solution

(A) $aL^2 + bL^3$ (B) $\frac{1}{2}\left(aL^2 + bL^3\right)$

(C) $\frac{aL^2}{2} + \frac{bL^3}{3}$ (D) $\frac{1}{2}\left(\frac{aL^2}{2} + \frac{bL^3}{3}\right)$

Q96. JEE Main 2013 (Online) 25th April Morning Slot

A wind-powered generator converts wind energy into electrical energy. Assume that the generator converts a fixed fraction of the wind energy intercepted by its blades into electrical energy. For wind speed v, the electrical power output will be most likely proportional to

Open solution

(A) v^4 (B) v^2 (C) v (D) v^3

Q97. JEE Main 2013 (Online) 23th April Morning Slot

A 70 kg man leaps vertically into the air from a crouching position. To take the leap the man pushes the ground with a constant force F to raise himself. The center of gravity rises by 0.5 m before he leaps. After the leap the c.g. rises by another 1 m. The maximum power delivered by the muscles is : (Take g = 10 ms^{-2})

Open solution

(A) 6.26×10^3 Watts at the start (B) 6.26×10^3 Watts at take off

(C) 6.26×10^4 Watts at the start (D) 6.26×10^4 Watts at take off

Q98. JEE Main 2013 (Online) 9th April Morning Slot

Two springs of force constants 300 N/m (Spring A) and 400 N/m (Spring B) are joined together in series. The combination is compressed by 8.75 cm. The ratio of energy stored in A and B is $\frac{E_A}{E_B}$. Then $\frac{E_A}{E_B}$ is equal to:

Open solution

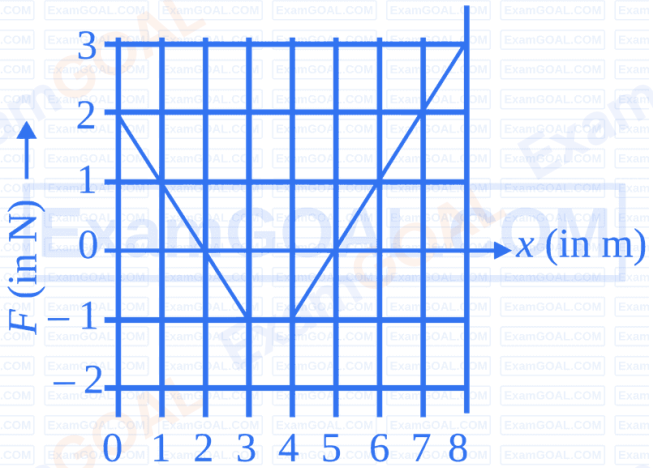
(A) $\frac{4}{3}$ (B) $\frac{16}{9}$ (C) $\frac{3}{4}$ (D) $\frac{9}{16}$

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Q99. AIEEE 2012 (Online) 26th May

Open solution

The force $\vec{F} = F\hat{i}$ on a particle of mass $2\ \text{kg}$, moving along the x-axis is given in the figure as a function of its position x. The particle is moving with a velocity of $5\ \text{m/s}$ along the x-axis at $x = 0$. What is the kinetic energy of the particle at $x = 8\ \text{m}$?



(A) $34\ \text{J}$ (B) $34.5\ \text{J}$ (C) $4.5\ \text{J}$ (D) $29.4\ \text{J}$

Q100. AIEEE 2012 (Online) 7th May

Open solution

A particle gets displaced by $\Delta\vec{r} = (2\hat{i} + 3\hat{j} + 4\hat{k})\ \text{m}$ under the action of a force $\vec{F} = (7\hat{i} + 4\hat{j} + 3\hat{k})$. The change in its kinetic energy is

(A) $38\ \text{J}$ (B) $70\ \text{J}$ (C) $52.5\ \text{J}$ (D) $126\ \text{J}$

Q101. AIEEE 2012

Open solution

This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.

If two springs S_1 and S_2 of force constants k_1 and k_2, respectively, are stretched by the same force, it is found that more work is done on spring S_1 than on spring S_2.

STATEMENT 1: If stretched by the same amount, work done on S_1 is more than that on S_2
STATEMENT 2: $k_1 < k_2$

(A) Statement 1 is false, Statement 2 is true
(B) Statement 1 is true, Statement 2 is false
(C) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation for Statement 1
(D) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation for Statement 1

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Q102. AIEEE 2010

The potential energy function for the force between two atoms in a diatomic molecule is approximately given by $U(x) = \frac{a}{x^{12}} - \frac{b}{x^6}$, where a and b are constants and x is the distance between the atoms. If the dissociation energy of the molecule is $D = [U(x = \infty) - U_{at\ equilibrium}]$, D is

(A) $\frac{b^2}{2a}$ (B) $\frac{b^2}{12a}$ (C) $\frac{b^2}{4a}$ (D) $\frac{b^2}{6a}$

Open solution

Q103. AIEEE 2008

An athlete in the Olympic games covers a distance of $100\ m$ in $10\ s$. His kinetic energy can be estimated to be in the range

(A) $200J - 500J$ (B) $2 \times 10^5 J - 3 \times 10^5 J$

(C) $20,000J - 50,000J$ (D) $2,000J - 5,000J$

Open solution

Q104. AIEEE 2007

A particle is projected at 60° to the horizontal with a kinetic energy K. The kinetic energy at the highest point is

(A) K/2 (B) K (C) Zero (D) K/4

Open solution

Q105. AIEEE 2007

A $2\ kg$ block slides on a horizontal floor with a speed of $4m/s$. It strikes an uncompressed spring and compresses it till the block is motionless. The kinetic friction force is $15N$ and spring constant is $10,000\ N/m$. The spring compresses by

(A) $8.5cm$ (B) $5.5cm$ (C) $2.5cm$ (D) $11.0cm$

Open solution

Q106. AIEEE 2006

A mass of $M\ kg$ is suspended by a weightless string. The horizontal force that is required to displace it until the string makes an angle of 45° with the initial vertical direction is

(A) $Mg\left(\sqrt{2}+1\right)$ (B) $Mg\sqrt{2}$ (C) $\frac{Mg}{\sqrt{2}}$ (D) $Mg\left(\sqrt{2}-1\right)$

Open solution

Q107. AIEEE 2006

A ball of mass $0.2\ kg$ is thrown vertically upwards by applying a force by hand. If the hand moves $0.2\ m$ while applying the force and the ball goes upto $2\ m$ height further, find the magnitude of the force. (consider $g = 10\,m/s^2$).

(A) $4N$ (B) $16\ N$ (C) $20\ N$ (D) $22\ N$

Open solution

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Q108. AIEEE 2006

A particle of mass $100g$ is thrown vertically upwards with a speed of $5\ m/s$. The work done by the force of gravity during the time the particle goes up is

(A) $-0.5J$ (B) $-1.25J$ (C) $1.25J$ (D) $0.5J$

Open solution

Q109. AIEEE 2006

The potential energy of a $1\ kg$ particle free to move along the x-axis is given by $V(x) = \left(\frac{x^4}{4} - \frac{x^2}{2}\right)J$. The total mechanical energy of the particle is $2J$. Then, the maximum speed (in m/s) is

(A) $\frac{3}{\sqrt{2}}$ (B) $\sqrt{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) 2

Open solution

Q110. AIEEE 2005

A bullet fired into a fixed target loses half of its velocity after penetrating $3\ cm$. How much further it will penetrate before coming to rest assuming that it faces constant resistance to motion?

(A) $2.0\ cm$ (B) $3.0\ cm$ (C) $1.0\ cm$ (D) $1.5\ cm$

Open solution

Q111. AIEEE 2005

The upper half of an inclined plane with inclination ϕ is perfectly smooth while the lower half is rough. A body starting from rest at the top will again come to rest at the bottom if the coefficient of friction for the lower half is given by

(A) $2\cos\ \phi$ (B) $2\,sin\ \phi$ (C) $\tan\ \phi$ (D) $2\tan\ \phi$

Open solution

Q112. AIEEE 2005

A spherical ball of mass $20\ kg$ is stationary at the top of a hill of height $100\ m$. It rolls down a smooth surface to the ground, then climbs up another hill of height $30\ m$ and finally rolls down to a horizontal base at a height of $20\ m$ above the ground. The velocity attained by the ball is

(A) $20\ m/s$ (B) $40\ m/s$ (C) $10\sqrt{30}\ m/s$ (D) $10\ m/s$

Open solution

Q113. AIEEE 2005

A body of mass m is accelerated uniformly from rest to a speed v in a time T. The instantaneous power delivered to the body as a function of time is given by

(A) $\frac{mv^2}{T^2}.t^2$ (B) $\frac{mv^2}{T^2}.t$
(C) $\frac{1}{2}\frac{mv^2}{T^2}.t^2$ (D) $\frac{1}{2}\frac{mv^2}{T^2}.t$

Open solution

Q114. AIEEE 2004

A particle moves in a straight line with retardation proportional to its displacement. Its loss of kinetic energy for any displacement x is proportional to

(A) x (B) e^x (C) x^2 (D) $\log_e x$

Open solution

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Q115. AIEEE 2004

A uniform chain of length $2\ m$ is kept on a table such that a length of $60\ cm$ hangs freely from the edge of the table. The total mass of the chain is $4\ kg$. What is the work done in pulling the entire chain on the table?

Open solution

(A) $12\ J$ (B) $3.6\ J$ (C) $7.2\ J$ (D) $1200\ J$

Q116. AIEEE 2004

A force $\vec{F} = \left(5\vec{i} + 3\vec{j} + 2\vec{k}\right) N$ is applied over a particle which displaces it from its origin to the point $\vec{r} = \left(2\vec{i} - \vec{j}\right) m$. The work done on the particle in joules is

Open solution

(A) $+10$ (B) $+7$ (C) -7 (D) $+13$

Q117. AIEEE 2004

A body of mass $'m'$, accelerates uniformly from rest to $'v_1'$ in time T. The instantaneous power delivered to the body as a function of time is given by

Open solution

(A) $\frac{mv_1t^2}{T}$ (B) $\frac{mv_1^2t}{T^2}$ (C) $\frac{mv_1t}{T}$ (D) $\frac{mv_1^2t}{T}$

Q118. AIEEE 2004

A particle is acted upon by a force of constant magnitude which is always perpendicular to the velocity of the particle, the motion of the particles takes place in a plane. It follows that

Open solution

(A) its kinetic energy is constant (B) is acceleration is constant

(C) its velocity is constant (D) it moves in a straight line

Q119. AIEEE 2003

A spring of spring constant $5 \times 10^3\ N/m$ is stretched initially by $5\ cm$ from the unstretched position. Then the work required to stretch it further by another $5\ cm$ is

Open solution

(A) $12.50\ N\text{-}m$ (B) $18.75\ N\text{-}m$ (C) $25.00\ N\text{-}m$ (D) $625\ N\text{-}m$

Q120. AIEEE 2003

A body is moved along a straight line by a machine delivering a constant power. The distance moved by the body in time $'t'$ is proportional to

Open solution

(A) $t^{3/4}$ (B) $t^{3/2}$ (C) $t^{1/4}$ (D) $t^{1/2}$

Q121. AIEEE 2002

A ball whose kinetic energy is E, is projected at an angle of 45° to the horizontal. The kinetic energy of the ball at the highest point of its flight will be

Open solution

(A) E (B) $\frac{E}{\sqrt{2}}$ (C) $\frac{E}{2}$ (D) zero

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Q122. AIEEE 2002

If a body loses half of its velocity on penetrating $3\ cm$ in a wooden block, then how much will it penetrate more before coming to rest?

(A) $1\ cm$ (B) $2\ cm$ (C) $3\ cm$ (D) $4\ cm$

Open solution

Q123. AIEEE 2002

A spring of force constant $800\ N/m$ has an extension of $5\ cm$. The work done in extending it from $5\ cm$ to $15\ cm$ is

(A) $16J$ (B) $8J$ (C) $32J$ (D) $24J$

Open solution

Numerical 42 questions

Q124. JEE Main 2026 (Online) 4th April Morning Shift

A 1 kg block subjected to two simultaneous forces $(2\hat{i}+3\hat{j}+4\hat{k})\mathrm{N}$ and $(3\hat{i}-\hat{j}-2\hat{k})\mathrm{N}$ is moved a distance of 25 m along $(3\hat{i}-4\hat{j})$ direction. The work done in this process is ____ J.

Open solution

Q125. JEE Main 2026 (Online) 4th April Morning Shift

A body of mass 2 kg begins to move under the influence of time dependent force $\vec{F}=\left(2t\hat{i}+6t^2\hat{j}\right)\mathrm{N}$, where $\hat{i}$ and $\hat{j}$ are unit vectors along x and y -axis respectively. The power produced by the force at $t=2\ \mathrm{s}$ is ____ W.

Open solution

Q126. JEE Main 2025 (Online) 23rd January Morning Shift

A force $\mathrm{f}=\mathrm{x}^2\mathrm{y}\hat{i}+\mathrm{y}^2\hat{j}$ acts on a particle in a plane $\mathrm{x}+\mathrm{y}=10$. The work done by this force during a displacement from $(0,0)$ to $(4\ \mathrm{m}, 2\ \mathrm{m})$ is _____ Joule (round off to the nearest integer)

Open solution

Q127. JEE Main 2024 (Online) 9th April Evening Shift

A force $(3x^2+2x-5)\mathrm{N}$ displaces a body from $x=2\ \mathrm{m}$ to $x=4\ \mathrm{m}$. Work done by this force is _____ J.

Open solution

Q128. JEE Main 2023 (Online) 15th April Morning Shift

A block of mass $10\ \mathrm{kg}$ is moving along x-axis under the action of force $F=5x\ N$. The work done by the force in moving the block from $x=2m$ to $4m$ will be _____ J.

Open solution

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Q129. JEE Main 2023 (Online) 13th April Evening Shift

A car accelerates from rest to $u\ \mathrm{m/s}$. The energy spent in this process is E J. The energy required to accelerate the car from $u\ \mathrm{m/s}$ to $2u\mathrm{m/s}$ is nE J. The value of n is _____.

Open solution

Q130. JEE Main 2023 (Online) 12th April Morning Shift

To maintain a speed of 80 km/h by a bus of mass 500 kg on a plane rough road for 4 km distance, the work done by the engine of the bus will be _____ kJ. [The coefficient of friction between tyre of bus and road is 0.04.]

Open solution

Q131. JEE Main 2023 (Online) 11th April Evening Shift

A block of mass $5\ \mathrm{kg}$ starting from rest is pulled up on a smooth inclined plane making an angle of 30° with horizontal with an effective acceleration of $1\ \mathrm{ms}^{-2}$. The power delivered by the pulling force at $t = 10\ \mathrm{s}$ from the start is _____ W.

[use $\mathrm{g} = 10\ \mathrm{ms}^{-2}$]

(calculate the nearest integer value)

Open solution

Q132. JEE Main 2023 (Online) 11th April Morning Shift

A force $\vec{F} = (2 + 3x)\hat{i}$ acts on a particle in the x direction where F is in newton and x is in meter. The work done by this force during a displacement from $x = 0$ to $x = 4\ \mathrm{m}$, is _____ J.

Open solution

Q133. JEE Main 2023 (Online) 10th April Evening Shift

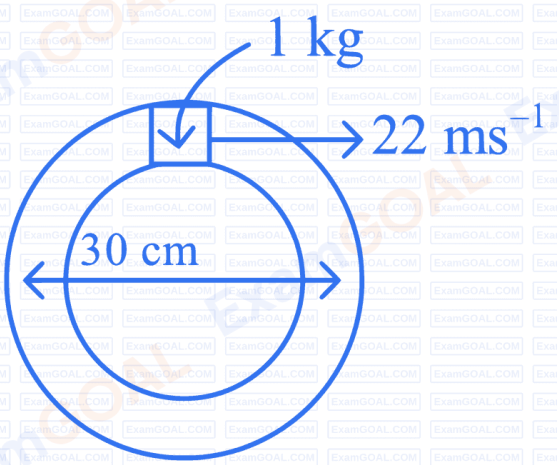
If the maximum load carried by an elevator is $1400\ \mathrm{kg}$ ($600\ \mathrm{kg}$ - Passengers + $800\ \mathrm{kg}$ - elevator), which is moving up with a uniform speed of $3\ \mathrm{m\,s}^{-1}$ and the frictional force acting on it is $2000\ \mathrm{N}$, then the maximum power used by the motor is _____ kW $\left(\mathrm{g} = 10\ \mathrm{m/s}^2\right)$

Open solution

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Q134. JEE Main 2023 (Online) 10th April Morning Shift

A closed circular tube of average radius 15 cm, whose inner walls are rough, is kept in vertical plane. A block of mass 1 kg just fits inside the tube. The speed of block is 22 m/s, when it is introduced at the top of tube. After completing five oscillations, the block stops at the bottom region of tube. The work done by the tube on the block is _____ J. (Given g = 10 m/s^2).



Open solution

Q135. JEE Main 2023 (Online) 8th April Evening Shift

A body of mass $5\ \mathrm{kg}$ is moving with a momentum of $10\ \mathrm{kg\ ms^{-1}}$. Now a force of $2\ \mathrm{N}$ acts on the body in the direction of its motion for $5\ \mathrm{s}$. The increase in the Kinetic energy of the body is _____ J.

Open solution

Q136. JEE Main 2023 (Online) 6th April Evening Shift

A body is dropped on ground from a height 'h_1' and after hitting the ground, it rebounds to a height 'h_2'. If the ratio of velocities of the body just before and after hitting ground is 4 , then percentage loss in kinetic energy of the body is $\frac{x}{4}$. The value of x is _____.

Open solution

Q137. JEE Main 2023 (Online) 6th April Morning Shift

A particle of mass $10\ \mathrm{g}$ moves in a straight line with retardation $2x$, where x is the displacement in SI units. Its loss of kinetic energy for above displacement is $\left(\frac{10}{x}\right)^{-n}$ J. The value of n will be _____

Open solution

Q138. JEE Main 2023 (Online) 1st February Evening Shift

A block is fastened to a horizontal spring. The block is pulled to a distance $x = 10\ \mathrm{cm}$ from its equilibrium position (at $x = 0$) on a frictionless surface from rest. The energy of the block at $x = 5$ cm is $0.25\ \mathrm{J}$. The spring constant of the spring is _____ $\mathrm{Nm^{-1}}$

Open solution

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Q139. JEE Main 2023 (Online) 1st February Evening Shift

A force $\mathrm{F} = \left(5 + 3y^2\right)$ acts on a particle in the y-direction, where F is in newton and y is in meter. The work done by the force during a displacement from $y = 2\ \mathrm{m}$ to $y = 5\ \mathrm{m}$ is _____ J.

Open solution

Q140. JEE Main 2023 (Online) 1st February Morning Shift

A small particle moves to position $5\hat{i} - 2\hat{j} + \hat{k}$ from its initial position $2\hat{i} + 3\hat{j} - 4\hat{k}$ under the action of force $5\hat{i} + 2\hat{j} + 7\hat{k}\ \mathrm{N}$. The value of work done will be _____ J.

Open solution

Q141. JEE Main 2023 (Online) 31st January Morning Shift

A lift of mass $\mathrm{M} = 500\ \mathrm{kg}$ is descending with speed of $2\ \mathrm{ms}^{-1}$. Its supporting cable begins to slip thus allowing it to fall with a constant acceleration of $2\ \mathrm{ms}^{-2}$. The kinetic energy of the lift at the end of fall through to a distance of $6\ \mathrm{m}$ will be _____ kJ.

Open solution

Q142. JEE Main 2023 (Online) 30th January Evening Shift

A body of mass $2\ \mathrm{kg}$ is initially at rest. It starts moving unidirectionally under the influence of a source of constant power P. Its displacement in $4\ \mathrm{s}$ is $\frac{1}{3}\alpha^2\sqrt{P}\ \mathrm{m}$. The value of α will be _____.

Open solution

Q143. JEE Main 2023 (Online) 29th January Morning Shift

A 0.4 kg mass takes 8s to reach ground when dropped from a certain height 'P' above surface of earth. The loss of potential energy in the last second of fall is _____ J.

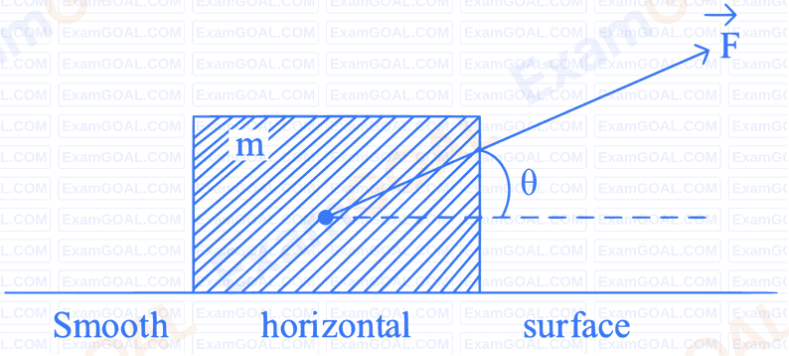
(Take g = 10 m/s^2)

Open solution

Q144. JEE Main 2023 (Online) 25th January Morning Shift

An object of mass 'm' initially at rest on a smooth horizontal plane starts moving under the action of force F = 2N. In the process of its linear motion, the angle θ (as shown in figure) between the direction of force and horizontal varies as $\theta = \mathrm{k}x$, where k is a constant and x is the distance covered by the object from its initial position. The expression of kinetic energy of the object will be $E = \frac{n}{k}\sin\theta$. The value of n is _____.

Open solution



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Q145. JEE Main 2023 (Online) 24th January Evening Shift

A body of mass 1kg begins to move under the action of a time dependent force $\vec{F} = \left(t\hat{i} + 3t^2\,\hat{j}\right)$ N, where $\hat{i}$ and $\hat{j}$ are the unit vectors along x and y axis. The power developed by above force, at the time t = 2s, will be _____ W.

Open solution

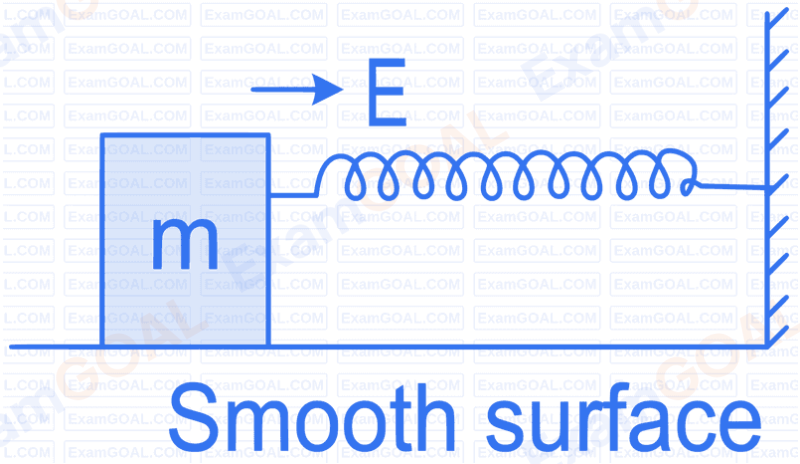
Q146. JEE Main 2023 (Online) 24th January Morning Shift

A spherical body of mass 2 kg starting from rest acquires a kinetic energy of 10000 J at the end of 5th second. The force acted on the body is _____ N.

Open solution

Q147. JEE Main 2022 (Online) 28th July Morning Shift

A block of mass 'm' (as shown in figure) moving with kinetic energy E compresses a spring through a distance $25\ \text{cm}$ when its speed is halved. The value of spring constant of used spring will be nE Nm^{-1} for $\text{n} =$ _____.



Open solution

Q148. JEE Main 2022 (Online) 25th June Morning Shift

A 0.5 kg block moving at a speed of 12 ms^{-1} compresses a spring through a distance 30 cm when its speed is halved. The spring constant of the spring will be _____ Nm^{-1}.

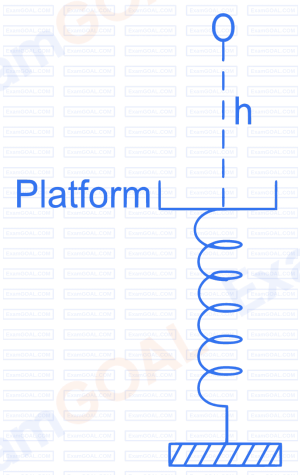
Open solution

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Q149. JEE Main 2022 (Online) 24th June Morning Shift

A ball of mass 100 g is dropped from a height h = 10 cm on a platform fixed at the top of a vertical spring (as shown in figure). The ball stays on the platform and the platform is depressed by a distance $\frac{h}{2}$. The spring constant is _____ Nm^{-1}.

(Use g = 10 ms^{-2})



Open solution

Q150. JEE Main 2021 (Online) 1st September Evening Shift

An engine is attached to a wagon through a shock absorber of length 1.5 m. The system with a total mass of 40,000 kg is moving with a speed of 72 kmh^{-1} when the brakes are applied to bring it to rest. In the process of the system being brought to rest, the spring of the shock absorber gets compressed by 1.0 m. If 90% of energy of the wagon is lost due to friction, the spring constant is _____ $\times 10^5$ N/m.

Open solution

Q151. JEE Main 2021 (Online) 31st August Morning Shift

A block moving horizontally on a smooth surface with a speed of 40 ms^{-1} splits into two equal parts. If one of the parts moves at 60 ms^{-1} in the same direction, then the fractional change in the kinetic energy will be x : 4 where x = _____.

Open solution

Q152. JEE Main 2021 (Online) 27th August Morning Shift

Two persons A and B perform same amount of work in moving a body through a certain distance d with application of forces acting at angle 45° and 60° with the direction of displacement respectively. The ratio of force applied by person A to the force applied by person B is $\frac{1}{\sqrt{x}}$. The value of x is ..... .

Open solution

Q153. JEE Main 2021 (Online) 26th August Morning Shift

A uniform chain of length 3 meter and mass 3 kg overhangs a smooth table with 2 meter lying on the table. If k is the kinetic energy of the chain in joule as it completely slips off the table, then the value of k is ..... . (Take g = 10 m/s^2)

Open solution

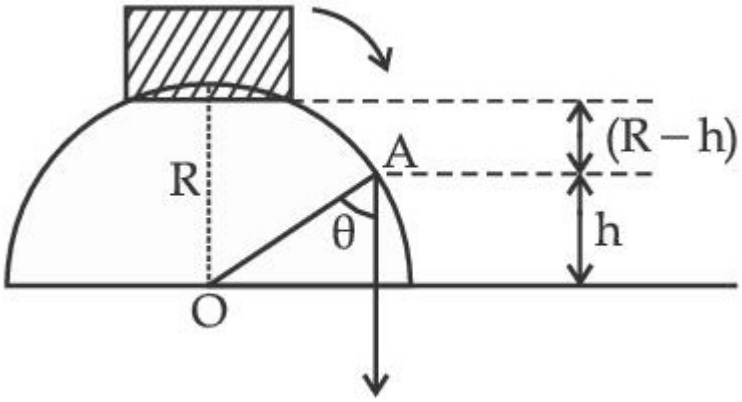
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Q154. JEE Main 2021 (Online) 27th July Evening Shift

A small block slides down from the top of hemisphere of radius R = 3 m as shown in the figure. The height 'h' at which the block will lose contact with the surface of the sphere is _____ m.

Open solution

(Assume there is no friction between the block and the hemisphere)



Q155. JEE Main 2021 (Online) 25th July Evening Shift

A force of $F = (5y + 20)\hat{j}$ N acts on a particle. The work done by this force when the particle is moved from y = 0 m to y = 10 m is _____ J.

Open solution

Q156. JEE Main 2021 (Online) 25th July Morning Shift

A pendulum bob has a speed of 3 m/s at its lowest position. The pendulum is 50 cm long. The speed of bob, when the length makes an angle of 60° to the vertical will be ($g = 10\ m/s^2$) _____ m/s.

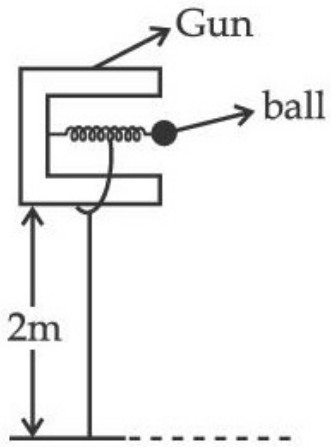
Open solution

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Q157. JEE Main 2021 (Online) 20th July Morning Shift

Open solution

In a spring gun having spring constant 100 N/m a small ball 'B' of mass 100 g is put in its barrel (as shown in figure) by compressing the spring through 0.05 m. There should be a box placed at a distance 'd' on the ground so that the ball falls in it. If the ball leaves the gun horizontally at a height of 2 m above the ground. The value of d is _____ m. ($g = 10\ m/s^2$).



Q158. JEE Main 2021 (Online) 18th March Evening Shift

Open solution

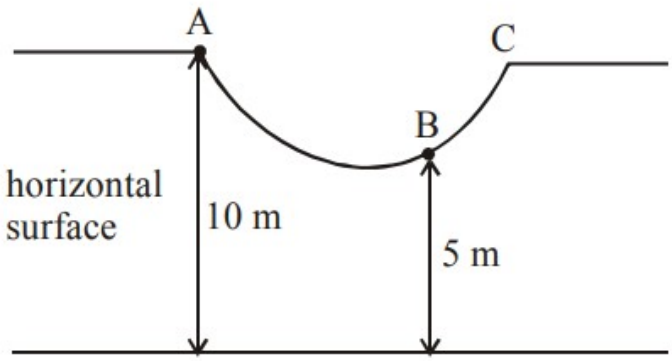
A ball of mass 4 kg, moving with a velocity of 10 ms^{-1}, collides with a spring of length 8 m and force constant 100 Nm^{-1}. The length of the compressed spring is x m. The value of x, to the nearest integer, is _____.

Q159. JEE Main 2021 (Online) 18th March Morning Shift

Open solution

As shown in the figure, a particle of mass 10 kg is placed at a point A. When the particle is slightly displaced to its right, it starts moving and reaches the point B. The speed of the particle at B is x m/s. (Take $g = 10\ m/s^2$)

The value of 'x' to the nearest integer is _____.



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Q160. JEE Main 2021 (Online) 25th February Morning Shift

The potential energy (U) of a diatomic molecule is a function dependent on r (interatomic distance) as

$$U = \frac{\alpha}{r^{10}} - \frac{\beta}{r^5} - 3$$

where, α and β are positive constants. The equilibrium distance between two atoms will be $\left(\frac{2\alpha}{\beta}\right)^{\frac{a}{b}}$, where a = _____.

Open solution

Q161. JEE Main 2020 (Online) 5th September Evening Slot

A body of mass 2 kg is driven by an engine delivering a constant power of 1 J/s. The body starts from rest and moves in a straight line. After 9 seconds, the body has moved a distance (in m) _____.

Open solution

Q162. JEE Main 2020 (Online) 3rd September Evening Slot

A block starts moving up an inclined plane of inclination 30° with an initial velocity of v_0 . It comes back to its initial position with velocity $\frac{v_0}{2}$. The value of the coefficient of kinetic friction between the block and the inclined plane is close to $\frac{I}{1000}$. The nearest integer to I is____.

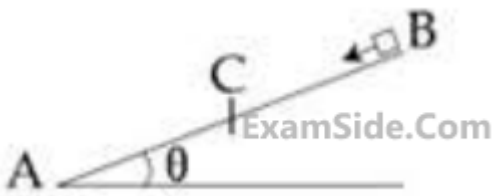
Open solution

Q163. JEE Main 2020 (Online) 3rd September Morning Slot

A cricket ball of mass 0.15 kg is thrown vertically up by a bowling machine so that it rises to a maximum height of 20 m after leaving the machine. If the part pushing the ball applies a constant force F on the ball and moves horizontally a distance of 0.2 m while launching the ball, the value of F (in N) is (g = 10 ms^{-2}) ____.

Open solution

Q164. JEE Main 2020 (Online) 2nd September Morning Slot



A small block starts slipping down from a point B on an inclined plane AB, which is making an angle θ with the horizontal section BC is smooth and the remaining section CA is rough with a coefficient of friction μ. It is found that the block comes to rest as it reaches the bottom (point A) of the inclined plane. If BC = 2AC, the coefficient of friction is given by μ = ktan θ . The value of k is _____.

Open solution

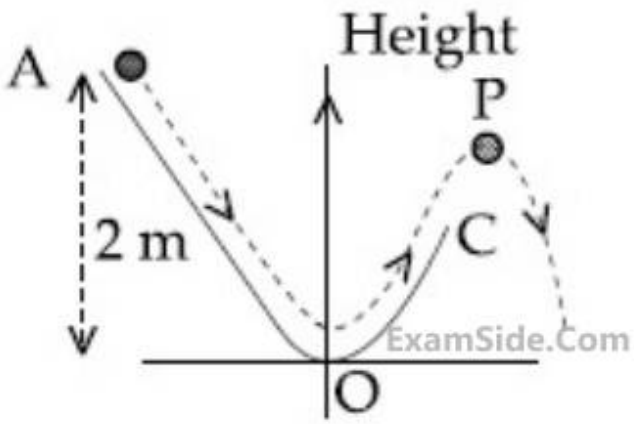
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Q165. JEE Main 2020 (Online) 7th January Morning Slot

Open solution

A particle (m = 1 kg) slides down a frictionless track (AOC) starting from rest at a point A (height 2 m). After reaching C, the particle continues to move freely in air as a projectile. When it reaches its highest point P (height 1 m), the kinetic energy of the particle (in J) is :

(Figure drawn is schematic and not to scale; take g = 10 ms^{-2})



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Answer Key

MCQ (Single Correct Answer)

1. **C**	2. **A**	3. **D**	4. **A**	5. **B**
6. **A**	7. **A**	8. **D**	9. **C**	10. **B**
11. **A**	12. **A**	13. **C**	14. **D**	15. **C**
16. **D**	17. **D**	18. **B**	19. **B**	20. **C**
21. **C**	22. **B**	23. **C**	24. **D**	25. **A**
26. **B**	27. **C**	28. **D**	29. **B**	30. **C**
31. **D**	32. **A**	33. **D**	34. **B**	35. **C**
36. **A**	37. **D**	38. **A**	39. **B**	40. **B**
41. **A**	42. **C**	43. **B**	44. **B**	45. **D**
46. **C**	47. **C**	48. **A**	49. **D**	50. **B**
51. **B**	52. **B**	53. **D**	54. **D**	55. **C**
56. **A**	57. **C**	58. **C**	59. **A**	60. **C**
61. **B**	62. **D**	63. **C**	64. **A**	65. **B**
66. **C**	67. **A**	68. **B**	69. **B**	70. **C**
71. **D**	72. **D**	73. **C**	74. **A**	75. **D**
76. **B**	77. **D**	78. **D**	79. **A**	80. **A**
81. **B**	82. **A**	83. **B**	84. **A**	85. **C**
86. **B**	87. **B**	88. **A**	89. **D**	90. **B**
91. **A**	92. **D**	93. **B**	94. **B**	95. **C**
96. **D**	97. **B**	98. **A**	99. **D**	100. **A**
101. **A**	102. **C**	103. **D**	104. **D**	105. **B**
106. **D**	107. **D**	108. **B**	109. **A**	110. **C**
111. **D**	112. **B**	113. **B**	114. **C**	115. **B**
116. **B**	117. **B**	118. **A**	119. **B**	120. **B**
121. **C**	122. **A**	123. **B**		

Numerical

124. **35**	125. **200**	126. **152**	127. **58**	128. **30**
129. **3**	130. **784**	131. **300**	132. **32**	133. **48**
134. **245**	135. **30**	136. **375**	137. **2**	138. **67**
139. **132**	140. **40**	141. **7**	142. **4**	143. **300**

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144. **2**	145. **100**	146. **40**	147. **24**	148. **600**
149. **120**	150. **16**	151. **1**	152. **2**	153. **40**
154. **2**	155. **450**	156. **2**	157. **1**	158. **6**
159. **10**	160. **1**	161. **18**	162. **346**	163. **150**
164. **3**	165. **10**			

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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